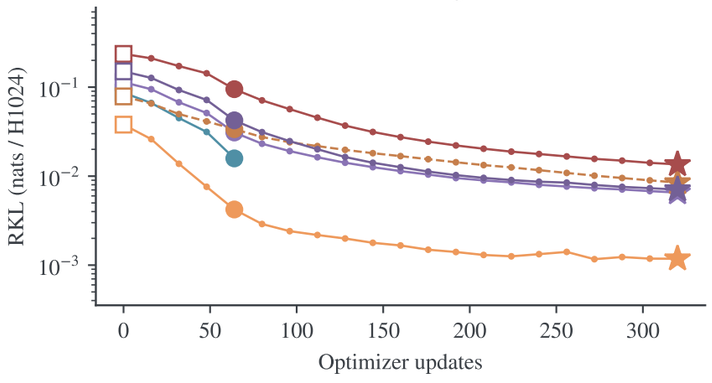
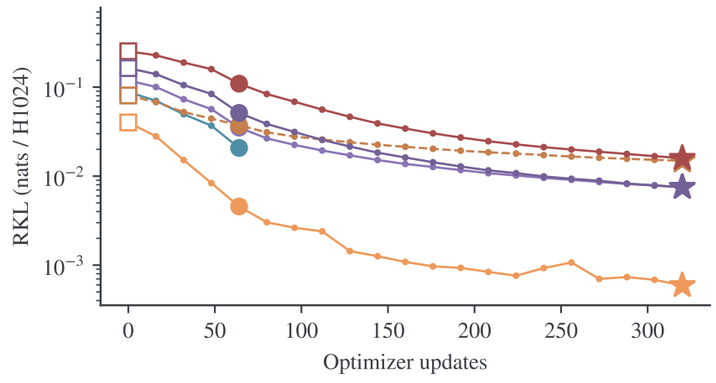
R1-1.5B | Fixed-state RKL vs. optimizer updates

Same size: solid = RL, dashed = added SFT, dotted = other controls

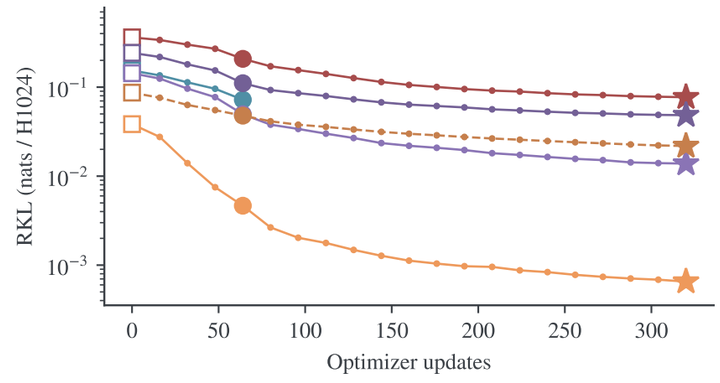
(a) Training



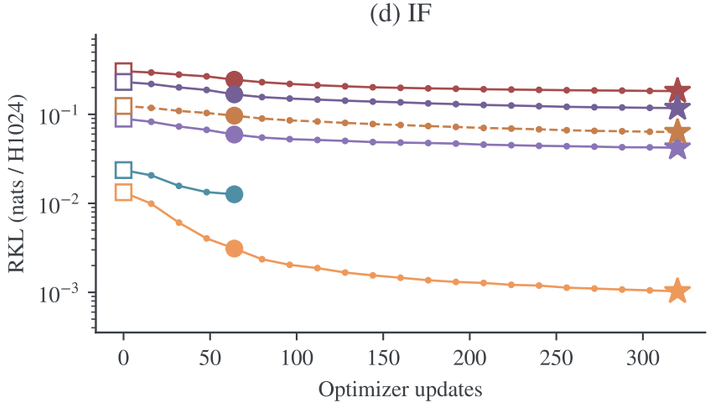
(b) Math



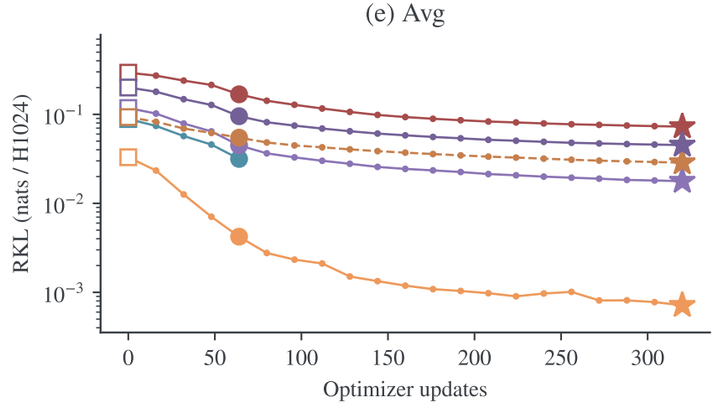
(c) Code



(d) IF



(e) Avg



deepcoder [64; RL]
deepscaler [320; RL]
justrl [320; RL]
minimath [320; SFT]
brorl [320; RL]
nemotron [320; RL]

Each dot = one snapshot, every 16 updates; square: 0; large circle: 64; star: 320.

Lines follow time within one run; snapshots are dependent, so no cross-run correlation is estimated.